

BHARGAV PATEL

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PROFESSIONAL SUMMARY

Computer Science graduate with hands-on experience in software development, algorithms, and databases. Proficient in C++, Python, Java, SQL, and modern web technologies. Skilled in building scalable applications, debugging complex systems, and collaborating in Agile teams. Passionate about applying problem-solving skills to deliver clean, efficient code that meets business needs. Eager to expand knowledge in areas like CI/CD, cloud technologies, and backend scalability while contributing to impactful engineering projects. Known for quickly adapting to new tools and mentoring peers through complex challenges.

EDUCATION

New Jersey Institute of Technology, Newark, NJ B.S. in Computer Science	GPA: 3.17/4.0	Jan 2023 - May 2025
Hudson County Community College, Jersey City, NJ A.S. in Computer Science	GPA: 3.78/4.0	Jan 2020 - Dec 2022

CERTIFICATIONS & ACHIEVEMENTS

- Unity Certified Developer – 3D & VR Development
- 3rd Place – Senior Capstone Project Showcase (VR Pedestrian Simulation)
- NSLS Leadership Certification
- Dean's List – Hudson County Community College

TECHNICAL SKILLS

Languages: C++, Python, Java, C, SQL, JavaScript, PHP, HTML/CSS, Bash

Frameworks & Tools: Git, GitHub, MongoDB, Unity (C#), scikit-learn, BeautifulSoup, Jupyter, VSCode

Databases: MySQL, MongoDB, PL/SQL

Concepts: Data Structures & Algorithms, Object-Oriented Programming, REST APIs, Test-Driven Development, Agile/Scrum, Debugging, Linux CLI

Other: Version Control, API Integration, Cloud Deployment (basic)

PROJECTS

Full-Stack Task Manager Web App | HTML, CSS/SCSS, JavaScript, PHP/MySQL

- Built a responsive task management system with frontend UI/UX and backend database integration.
- Implemented secure login, CRUD functionality, and desktop-optimized layouts.
- Styled UI with SCSS variables and mixins for maintainable, scalable code.
- While designing the UI/UX, became motivated to explore CI/CD pipelines to automate deployments.
- Strengthened full-stack development skills by managing both client-side design and server-side logic.

Pricing & Features Website (Multi-Page, SCSS-Driven Design) | HTML, SCSS

- Developed a 5-page responsive website with sections for features, pricing, contact, and premium.
- Enhanced UI/UX with reusable SCSS mixins, grid/flexbox layouts, and styled tooltips.
- Designed visually consistent page bottoms to avoid blank sections and improve user engagement.
- Practiced responsive design principles to ensure accessibility across desktop and mobile.

VR Pedestrian Safety Simulation (Capstone – 3rd Place Showcase) | Unity, C#

- Led a 4-member team to design a VR simulation for pedestrian safety with KATVR treadmill hardware.
- Integrated immersive interactions, backend logic, and hardware/software synchronization.
- Earned 3rd place showcase award for technical innovation and presentation impact.
- Learned to manage project deadlines, task delegation, and cross-functional collaboration.

AI Game Bot – Connect Four Solver | Python

- Developed an AI player using minimax with alpha-beta pruning, reducing computation time by ~40%.
- Designed state-space optimization strategies to ensure optimal gameplay.
- Strengthened knowledge of algorithms, data structures, and performance tuning.
- Demonstrated ability to balance theoretical computer science with practical implementation.

Compiler Construction Projects | C++

- Built a lexical analyzer, parser, and interpreter for a Pascal-like language.
- Implemented recursive-descent parsing, semantic checks, and runtime error handling.
- Gained experience in system-level programming, debugging, and compiler design.
- Improved understanding of how programming languages work from syntax to execution.

Cybersecurity Lab Project – Penetration Testing & IDS | Linux, Nmap, Snort, Security Onion

- Simulated attacks (port scanning, DoS, exploitation) using Nmap, Nessus, Metasploit.
- Configured IDS with Snort rules, Security Onion, and SGUIL for threat detection.
- Practiced penetration testing, security reporting, and incident response workflows.
- Gained appreciation for secure coding practices and defensive software engineering.

Machine Learning Projects | Python, scikit-learn, Pandas, Matplotlib

- Credit Card Categorization: Classified scores using Decision Tree, KNN, Naive Bayes.
- Weather Predictor: Built regression models with feature scaling and interpretability.
- Salary Prediction: Predicted income levels using Random Forest ensembles.
- Applied data preprocessing, model evaluation, and visualization techniques.
- Strengthened data-driven problem solving and applied ML concepts to real-world datasets.

Algorithms & Optimization | Python

- Solved the 15-puzzle problem with an efficient A* search algorithm.
- Implemented Huffman coding for encoding, decoding, and compression.
- Practiced dynamic programming and algorithmic problem-solving in academic projects.
- Reinforced critical thinking and efficiency analysis in algorithm design.

Data Scraping & Analysis | Python, BeautifulSoup, Pandas, Matplotlib

- Automated web scraping of raw HTML datasets using BeautifulSoup.
- Cleaned and structured data, creating visualizations for pattern analysis.
- Combined backend scripting with frontend reporting to generate insights.
- Practiced end-to-end workflow from raw data collection to business-ready results.

3D Unity Game Development | Unity, C#

- Built multiple interactive 3D games featuring physics, animations, and custom logic.
- Designed playable levels with attention to UI/UX and immersive user experience.
- Earned Unity Certification, validating proficiency in 3D and VR development.
- Applied creative problem-solving to deliver engaging gameplay and smooth mechanics.

EXPERIENCE

Hudson County Community College *Enrollment Assistant*

Jersey City, NJ, USA *Mar 2022 - June 2022*

- Helped 100+ students with admissions & transcript issues; maintained accurate records.
- Delivered clear communication across in-person and phone interactions.
- Supported administrators in resolving enrollment bottlenecks and improving service.